

■ PE better
 ■ no difference
 ■ Standard better
 outlined: 95% CI excludes 0

Rotated shifted Rastrigin

Dimension d	1/2	2/3	5/6	1	7/6	4/3	3/2
2	+0.81	+0.61	-0.10	-0.30	+0.39	+0.20	+0.28
5	+0.03	-0.13	-0.22	+0.15	-0.57	-0.48	+0.00
10	+0.59	+0.69	-0.60	-0.30	+0.32	+0.01	-0.15
20	+3.40	+0.99	-0.09	-4.03	-2.60	-3.59	-1.85
30	+1.89	+1.95	+2.84	+2.57	+2.74	+0.44	+1.48
	Decay exponent α						

Rosenbrock

Dimension d	1/2	2/3	5/6	1	7/6	4/3	3/2
2	-0.007	-0.002	-0.010	-0.009	-0.003	-0.003	-0.003
5	-0.027	-0.018	+0.000	-0.002	+0.005	-0.003	-0.006
10	+0.011	+0.030	+0.010	-0.001	-0.013	-0.000	-0.000
20	+0.153	+0.151	+0.150	+0.132	+0.115	+0.113	+0.096
30	+0.241	+0.222	+0.257	+0.235	+0.200	+0.096	+0.073
	Decay exponent α						

Shifted Ackley

Dimension d	1/2	2/3	5/6	1	7/6	4/3	3/2
2	+0.04	+0.01	-0.50	-0.02	-0.49	-0.51	-0.01
5	+0.14	-0.66	+0.25	-1.29	-0.27	-0.11	+0.19
10	+0.56	+0.10	+0.16	+0.25	+0.47	+0.49	+0.04
20	+0.09	+0.36	-0.05	+0.36	-0.24	-0.05	+0.35
30	+1.30	+1.18	+1.40	+1.59	+0.55	+0.90	+1.16
	Decay exponent α						

Cells show PE improvement over Standard for Decay Uniform under GP-UCB, 400 BO steps, 5 paired seeds. Colour is scaled symmetrically about zero, independently per benchmark.